THE UNIVERSITY OF NAIROBI

FACULTY OF ENGINEERING

DEPARTMENT OF ELECTRICAL AND INFORMATION ENGINEERING

FINAL YEAR PROJECT REPORT

PROJECT INDEX: PRJ 010

DESIGN OF AN ELECTRONIC BIRD DETERRENT SYSTEM FOR SMART RICE FARMING

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This project report is submitted in partial fulfilment of the requirement for the award of the Degree of Bachelor of Science in Electrical and Electronics Engineering from the University of Nairobi

Submitted on 4th of June 2025

DECLARATION OF ORIGINALITY

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DEDICATION

I dedicate this work to all the publishers, scholars, journalists, and members of the academic community who tirelessly share their work with the world. It is through their generosity and commitment to sharing knowledge that we learn from one another and continue to advance our collective understanding. May this project stand as a contribution to that noble mission.

ACKNOWLEDGEMENTS

This project forms part of the AFRETEC initiative and was developed in collaboration with Carnegie Mellon University Africa, the University of Rwanda, and the University of Nairobi. I am therefore profoundly grateful to AFRETEC for financing and overseeing the project, Dr. Cosmas Raymond Mutugi Kiruki for his guidance and leadership throughout the course of the project, the University of Nairobi for facilitating the operations of the project, and the collaborating teams at Carnegie Mellon University Africa, the University of Rwanda, and the University of Nairobi for their collective efforts in turning this project into a reality.

TABLE OF CONTENTS

DECLARATION OF ORIGINALITY 2
DEDICATION 3
ACKNOWLEDGEMENTS 4
TABLE OF CONTENTS 5
LIST OF TABLES 8
LIST OF FIGURES 9
LIST OF ABBREVIATIONS 11
ABSTRACT 12
1 INTRODUCTION 13
1.1 Background 13
1.2 Objectives 15
1.2.1 Main Objective 15
1.2.2 Specific Objectives 15
1.3 Scope of Work 15
2 LITERATURE REVIEW 16
2.1 Approaches to Bird-Deterring 16
2.1.1 Chemical and Pesticide-Based Bird Repellents 17
2.1.2 Visual Deterrents 17
2.1.3 Bioacoustics Deterrents 17
2.1.4 Hired Bird Scarers 18
2.1.5 Electronic Bird Deterrents 18
2.2 Sensor-Actuator Systems in Bird Deterrents Technologies 19
2.2.1 PIR Sensor 19
2.2.2 Ultrasonic Sensors 19
2.2.3 Camera 20
2.2.4 Sound Sensors/Microphone 21
2.2.5 Laser Deterrents 22
2.2.6 Ultrasonic Deterrents 23
2.2.7 Piezo Buzzer 24
2.2.8 MP3 Module and Speaker 25
2.2.9 Servo Motor 25

2.2.10 Drones 26
2.2.11 Motion Activated Jet Spray 27
2.2.12 Multi-Actuation Approach 28
2.3 Comparative Analysis of Birds Deterrents Actuation Methods 28
2.3.1 Birds Deterrents Comparative Analysis Table 29
2.4 Top 3 Bird Deterrent Options 30
2.4.1 Visual Deterrents (Laser and Servo Motors) 30
2.4.2 Bioacoustics (Audio Stimuli) 30
2.4.3 Water Jet Spray System 30
2.4.4 Synergy with Drones 31
2.5 Factors affecting effectiveness 31
2.6 Cost-Effectiveness Considerations 32
3 DESIGN METHODOLOGY 33
3.1 Visual Repellants 33
3.2 Bioacoustics Repellants 34
3.3 The Controller 36
3.3.1 Microcontroller Requirements for Bird Repellent System 36
3.4 Bird Repellent System Block Diagram 38
3.5 Audio Stimuli 38
3.6 Visual Stimuli 40
3.7 System Flowchart 40
3.8 Simulation on Wokwi 41
3.9 Prototype 42
3.9.1 Breadboard 42
3.9.2 Electrical Hardware Design 42
3.9.3 Key components 43
4 IMPLEMENTATION 45
4.1 Electrical Hardware Implementation 45
4.1.1 Core Features 46
4.2 Software Implementation 47
4.2.1 Preliminaries 47
4.2.2 Installing CP210x USB to UART Bridge VCP Drivers (Windows) 48

4.2.3 Arduino IDE 48
4.3 Program Implementation 49
5 RESULTS 51
6 RECOMMENDATIONS 53
7 CONCLUSION 54
8 REFERENCES 55
9 APPENDICES 59
9.1 APPENDIX 1: Bill of Quantities 59
9.2 Appendix 2: BirdRepellent.ino 60
9.3 Appendix 3: servoControl.h 61
9.4 Appendix 4: servoControl.cpp 61
9.5 Appendix 5: slideSwitch.h 65
9.6 Appendix 6: slideSwitch.cpp 65
9.7 Appendix 7: DFPlayerControl.h 67
9.8 Appendix 8: DFPlayerControl.cpp 68

LIST OF TABLES

Table 1: Birds Repellent Comparative Analysis..... 29

LIST OF FIGURES

Figure 1: Sustainable Development Goals 13
Figure 2: Quelea birds are destroying rice, sorghum, millet and wheat fields 14
Figure 3: Scarecrows 17
Figure 4: A man scares away destructive quelea birds at a farm in Ahero 18
Figure 5: A Proposed Design utilizing IoT 19
Figure 6: PIR Motion Sensor 19
Figure 7:HC-SR04 ultrasonic sensor 20
Figure 8: Raspberry Pi Camera V2 8MP 21
Figure 9: Sound Sensor Module 22
Figure 10: Birds agitated by the green laser beam 22
Figure 11: Concept of the proposed automatic laser bird-repellent system 23
Figure 12: PLT5 510 from OSRAM Opto Semiconductors 23
Figure 13: Bird's reaction before and after being exposed to a 60 KHz frequency 24
Figure 14: Piezo Buzzer 24
Figure 15: TowerPro SG 90 Micro Servo Motor 25
Figure 16: DJI Mini 3 Pro Quadcopter Drone with RC-N1 Remote Controller 26
Figure 17: System Design 26
Figure 18: RobotFalcon's view as it hunts 27
Figure 19: Jet spray deters the bird with a burst of water 27
Figure 20: Sound and Motion Output Model of Scarecrow 28
Figure 21: A laser repellent system 34
Figure 22:Pan and Tilt with Micro Servo 34
Figure 23: The DFPlayer Mini MP3 Player and a SanDisk 64 GB SD Card 35
Figure 24:DFPlayer Mini Pin Map 35
Figure 25: A 2 Watts 8 Ohm MP3 Speaker 36
Figure 26: ESP-WROOM-32 Dual Core CPU with Low Power Consumption 37
Figure 27: ESP-WROOM-32 pinout 37
Figure 28: Block Diagram Representation of the System 38
Figure 29: BirdWeather - a living library of bird vocalization 39
Figure 30: Some Natural Bird' Sounds in BirdWeather 39
Figure 31: Laser Bird Control Device with Laser Light to Repel Bird 40
Figure 32: General flowchart of the bird deterrent algorithm 41
Figure 33: Simulation on Wokwi (Web-Based Simulator) 41
Figure 34: Breadboard version 42
Figure 35: Actuation Board Schematic 43
Figure 36: Slide Switch SPDT 44
Figure 37: The Actuation PCB Top View 45
Figure 38: The Actuation PCB Bottom View 45
Figure 39: The Actuation PCB 46
Figure 40: 20 MP3 sound files stored on the SD card for playback when birds are detected. 46
Figure 41: Prototype 47

Figure 42: Additional Boards Manager 49
Figure 43:Arduino Code Split into .h and .cpp Files for Modular Design 50
Figure 44: Bird Repellent Project Files 50
Figure 45: Arduino IDE's Serial Monitor..... 51

LIST OF ABBREVIATIONS

GPS	Global Positioning System
UART	Universal asynchronous receiver-transmitter
SPI	Serial Peripheral Interface
IDE	Integrated Development Environment
PWM	Pulse Width Modulation
LASER	Light Amplification by Stimulated Emission of Radiation
PIR	Passive Infrared Sensor
PCB	Printed Circuit Board
MCU	Microcontroller Unit
IoT	Internet of Things
API	Application Programming Interface
GPIO	General Purpose Input/Output
DC	Direct Current
SERVO	Servomechanism
UAV	Unmanned Aerial Vehicle
I^2C	Inter-Integrated Circuit
FDA	Food and Drug Administration
CMU	Carnegie Mellon University
SDG	Sustainable Development Goal
IEC	International Electrotechnical Commission

ABSTRACT

According to the **United Nations Food and Agriculture Organization** (FAO), a single quelea bird can consume approximately 10 grams of grain daily. In large flocks numbering around 2 million, these birds can collectively devour up to 20 tons of grain within the same period [1]

Bird damage poses a major challenge for rice farmers globally, leading to significant economic losses through decreased yields and compromised crop quality. Birds primarily attack rice fields during two critical phases: right after seeding, when they consume newly planted seeds, and during the ripening stage, when they feed on mature grains.

Traditional bird deterrent methods—such as scarecrows, reflective tapes, and manual noise-making—have become increasingly ineffective as birds quickly adapt to these static measures. This limitation has driven the development of advanced electronic deterrent systems that utilize technology to offer dynamic and adaptive solutions.

This project designs the hardware and control algorithms for an electronic bird repellent's actuation subsystem (lasers, servos, bioacoustics), which interfaces with a separate AI-based detection module developed by Carnegie Mellon University.

By addressing bird-related crop losses, this project report supports the United Nations Sustainable Development Goal **(SDG) 2: Zero Hunger**, which aims to end hunger, achieve food security, and promote sustainable agriculture. The adoption of innovative and eco-friendly bird deterrent technologies contributes to increasing rice yields sustainably, enhancing food security for farming communities worldwide [2].

1 INTRODUCTION

1.1 Background

The global SDG Goals 2 and 12 emphasize the importance of ending hunger, achieving food security, promoting sustainable agriculture (SDG 2), and ensuring responsible consumption and production patterns (SDG 12). This project aligns with these goals by supporting sustainable rice farming practices that enhance local food production, reduce crop losses, and minimize environmental impact.

Figure 1: Sustainable Development Goals

Rice farming remains an important concern in Kenya due to its positive impact on increasing household food security, raising farmer's income as well as reducing risks in the years of poor weather conditions. Currently, the demand for rice in Kenya outstrips its production, a gap that is filled through imports [3].

Africa's inability to reach self-sufficiency in rice is the result of several major constraints in the rice industry which require urgent redress to stem the trend of over-reliance on imports and to satisfy the increasing demand for rice in areas where the potential of local production resources is exploited at very low levels [4].

Birds can cause significant economic damage to rice farms through:

i Consumption of newly planted seeds, reducing plant population.
ii Pulling of newly germinated seedlings.
iii Feeding on ripening grain.
iv Creating bare patches in fields that reduce overall yield.
v Decreasing grain quality through partial consumption.

The most problematic bird species for rice farms include red-winged blackbirds, grackles, quelea birds, brown-headed cowbirds, sparrows, and various waterfowl. These birds can devastate large areas of rice fields in remarkably short periods, with some farmers reporting complete destruction of newly seeded fields within hours [1].

Figure 2: Quelea birds are destroying rice, sorghum, millet and wheat fields

Characterized by large numbers, quelea birds are known to leave a trail of destruction behind them as the super colonies sweep through rice fields [5].

1.2 Objectives

1.2.1 Main Objective

To investigate and design the hardware and software for the actuation system of a bird repellent device that aims at deterring birds in a rice farm environment.

1.2.2 Specific Objectives

a. To review the literature on bird repellent systems, focusing on deterrent mechanisms.
b. To design and implement hardware and software for the actuation subsystem.
c. To design interfaces for seamless integration with detection subsystem.

1.3 Scope of Work

The specific objectives will be achieved through the following tasks:

Task 1: Conduct a comprehensive review of the existing literature on bird repellent systems, focusing on the various deterrent mechanisms.
Task 2: Design the hardware components of the actuation system
Task 3: Develop firmware and control algorithms for real-time activation of the actuators based on sensor feedback from bird detection modules.
Task 4: Integrate the other subsystems with the actuation system.

2 LITERATURE REVIEW

2.1 Approaches to Bird-Deterring

Birds are a huge problem in agriculture due to their insatiable appetite for farm produce. Various deterrent strategies have been developed, combining traditional and modern technologies. Automated bird-repellent systems employ sensors and actuators, providing a sustainable and innovative solution for bird deterrence [6].

Ornithologists, who specialize in the study of birds, contribute valuable insights into bird behavior, aiding engineers in the design of effective and humane deterrent methods.

The difficulty of defining or measuring intelligence in non-human animals makes the subject of intelligence difficult to study scientifically in birds [7]. The corvids (ravens, crows, jays, magpies, etc.) and parrots are often considered the most intelligent birds, and are among the most intelligent animals in general. Pigeons, finches, chickens, and birds of prey have also been common subjects of intelligence studies [7]. The figures below depict examples that highlight the cognitive abilities of birds.

Eastern carrion crows (*Corvus corone orientalis*) in Japan have been observed taking advantage of road traffic to crack hard-shelled nuts.

A woodpecker finch using a stick to impale a grub, with a second image showing it had successfully captured it

Birds communicate using visual signals as well as through the use of calls and song. The testing of intelligence in birds is therefore usually based on studying responses to sensory stimuli [7].

2.1.1 Chemical and Pesticide-Based Bird Repellents

Chemical repellents have been commonly used to deter birds by making crops and farms unattractive or toxic. However, their long-term environmental impact raises concerns [6].

Limitations include potential harm to birds and pollution of ecosystems, which underscores the need for more eco-friendly solutions.

"We need to look at the ecosystem as a network more like a food chain. When you eradicate one species completely, they will be a knockoff effect in the web that exists within the ecosystem. I don't think it's a wise move on eradicating a species" said a program manager at Nature Kenya when the County Government of Kisumu planned to eliminate 5.8 million quelea birds ravaging Ahero rice farms [1].

2.1.2 Visual Deterrents

Scarecrows are the most well-known visual deterrents, dating back centuries. These objects are designed to resemble predators (e.g., humans or large birds) to scare birds away from agricultural fields.

Studies suggest that visual deterrents like scarecrows shown in Figure 3 are more effective when combined with audible deterrents and/or devices capable of physical movement, increasing the perceived threat to birds [8].

Figure 3: Scarecrows

2.1.3 Bioacoustics Deterrents

Animals produce sounds for self-defense, territory marking, mate attraction, social communication, and navigation. Ethologists focus on recording and analyzing these vocalizations to understand their functions and significance [9].

Audio deterrents emit high-frequency sound waves or avian distress calls to drive birds away. Birds are biologically wired to react to distress signals, which mimic danger. Historically, the application of sound to deter vermin dates back to ancient Chinese practices [6].

However, the birds' ability to adapt to repeated sounds is a concern. This calls for dynamic systems that vary stimuli, such as integrating sound with motion or light, which can be achieved through sensor-actuated responses.

2.1.4 Hired Bird Scarers

Consultations with farmers facing avian challenges on rice farms revealed that manual bird deterrence is labor-intensive and often includes shouting, clapping, waving, and escalating measures, such as throwing stones or dirt at birds. Despite these laborious efforts, birds retreat to nearby perches and monitor the situation before resuming their destructive activities. This underscores the need for a **multi-actuation and dynamic approach** to ensure sustained deterrence.

Figure 4: A man scares away destructive quelea birds at a farm in Ahero

2.1.5 Electronic Bird Deterrents

Electronic bird-repellent devices offer a cost-effective, scalable, and promising long-term solution. Unlike chemical repellents, electronic devices do not release harmful pollutants.

The electronic bird-repellent system can send notifications to the user's mobile device when birds are detected, allowing real-time monitoring as demonstrated in Figure 5 [10].

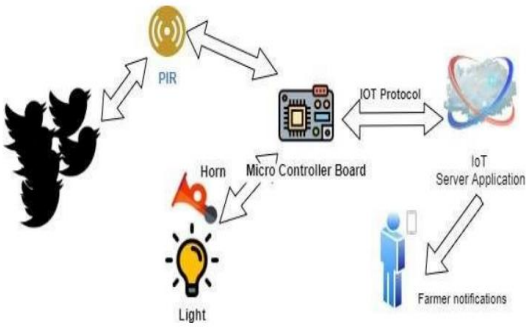


Figure 5: A Proposed Design utilizing IoT

2.2 Sensor-Actuator Systems in Bird Deterrents Technologies

2.2.1 PIR Sensor

Figure 6 shows a passive infrared (PIR) sensor that detects infrared radiation emitted by objects in its view. PIR sensors are commonly used in motion detectors, sensing changes in IR radiation to identify movement. PIR (Passive Infrared) sensors detect motion, particularly when birds enter the protected area [10]. The PIR sensor has an effective detection distance of around five meters with an area of around seven meters [11].

Figure 6: PIR Motion Sensor

2.2.2 Ultrasonic Sensors

Ultrasonic sensors measure distance by emitting ultrasonic sound waves and detecting the time it takes for the sound to bounce back from an object. Figure 7 shows a HC-SRO4 ultrasonic sensor.

These sensors typically consist of a transmitter that emits the sound and a receiver that picks up the echo. The time delay between transmission and reception is crucial in calculating the distance to the object using the formula:

Distance = Speed of Sound in Air * (Time Taken / 2)

In bird-repellent systems, ultrasonic sensors can detect birds approaching and trigger deterrents [12], making them effective in diverse conditions.

Figure 7:HC-SR04 ultrasonic sensor

2.2.3 Camera

An AI-based bird detection project should have a camera that provides accurate and real-time identification. It should capture high-quality images in various lighting and at decent distances.

Deep learning will function as a tool to classify birds based on the images captured by the camera [13]. Key features of the camera should include:

1. **Resolution**: Good resolution for detailed images, ensuring small or distant birds can be detected clearly.
2. **Frame Rate**: Good FPS or higher to capture smooth motion and avoid missing bird activity.
3. **Field of View**: A wide-angle lens to cover a wider area with fewer cameras.
4. **Low Light Performance**: Infrared (IR) or night vision for effective detection during low-light conditions.
5. **Weatherproofing**: Protection against rain, dust, and temperature changes.
6. **Connectivity**: Reliable data transmission to the processing unit, especially in remote locations.

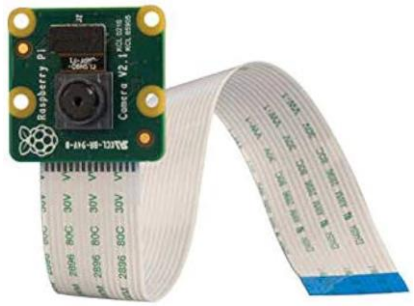


Figure 8: Raspberry Pi Camera V2 8MP

2.2.4 Sound Sensors/Microphone

A sound sensor is a module that detects sound waves via the sound's intensity and then converts it to electric signals. They are used to detect bird sounds, helping identify their presence [14]. The sound sensor should have the following features:

1. **Sound Detection**: The sensor can detect surrounding sounds and output an analog voltage value corresponding to sound intensity.
2. **Adjustable Sensitivity**: It features adjustable sensitivity settings to tailor the sensor's responsiveness to various sound levels, ensuring stable performance in different environments.
3. **High Sensitivity with Amplifier**: The built-in amplifier circuit enhances sensitivity, allowing detection of quieter sounds. The gain can be adjusted to fine-tune the sensor's response.
4. **Analog Output with AD Conversion**: The sensor provides an analog output that can be transformed into a digital signal using an Analog-to-Digital Converter (ADC). This allows for precise measurement and processing.

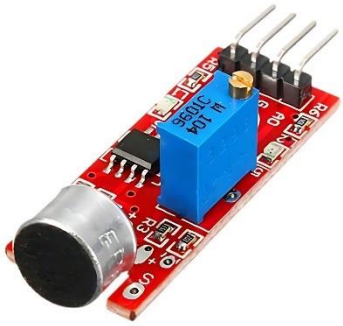


Figure 9: Sound Sensor Module

2.2.5 Laser Deterrents

Light Amplification by Stimulated Emission of Radiation (Laser) is a device that produces a narrow, focused beam of light through optical amplification.

As a bird-repellent actuator, they are non-lethal, humane, and environmentally friendly, as they do not rely on chemicals or cause physical harm. They are particularly effective in low-light conditions and can cover large areas.

Figure 10: Birds agitated by the green laser beam

Green lasers are especially successful, as most birds' visual systems perceive green light more intensely than other colors—about eight times more than red. This makes them highly sensitive to green beams, which they construe as predators, triggering a flight response.

The laser's sweep pattern is also critical; companies like Bird Control Group have refined the patterns to optimize bird dispersion, ensuring consistent effectiveness [15].

When a wild bird appears at a farm, the proposed system in Figure 11 detects the bird's position in an image captured by its detection unit and then uses a laser beam to repel the bird [16].

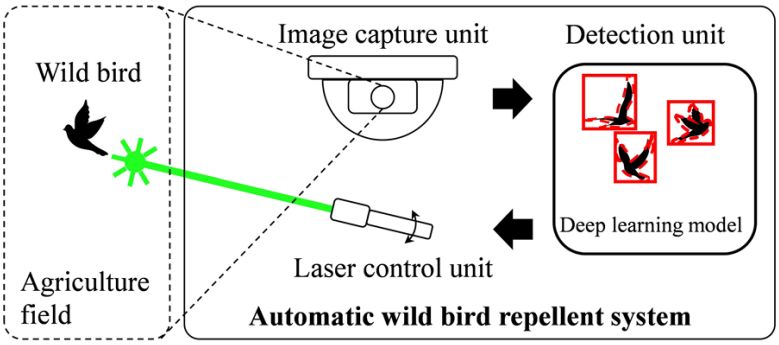


Figure 11: Concept of the proposed automatic laser bird-repellent system

Laser diodes can produce a narrow beam of laser light in which all the light waves have the same wavelength (monochromatic) and are coherent. This allows laser beams to be very bright and focused over a tiny spot. The PLT5 510 shown in Figure 12 from OSRAM Opto Semiconductors is an affordable 10 mW laser diode.

Figure 12: PLT5 510 from OSRAM Opto Semiconductors

2.2.6 Ultrasonic Deterrents

Ultrasonic waves are sound waves with frequencies higher than the upper limit of human hearing, typically above 20 kHz. Some birds are sensitive to ultrasonic waves. In Figure 13 a bird is exposed to an ultrasonic 60 KHz wave from an LC oscillator circuit [17].

An ultrasonic bird-repellant system would require an ultrasonic speaker and circuit to produce and amplify the ultrasonic waves. A 555 timer IC can be used in Astable Mode to generate ultrasonic sound waves. The frequency of oscillation is given by:

$$F = 1.44 / [(R1+2\times R2) \times C1] \text{ Hz}$$

However, ultrasonic sounds may deter certain species but will not repel other avian species with similar audio ranges as humans. Ultrasonic waves may deter certain animals such as cats and dogs, but their effect on birds is minimal [18].

Figure 13: Bird's reaction before and after being exposed to a 60 KHz frequency

2.2.7 Piezo Buzzer

A **piezo buzzer** is an electronic device that uses the piezoelectric effect to generate sound at a low power consumption. It operates by applying an alternating electrical signal to a piezoelectric ceramic disk, which causes the disk to vibrate and produce sound waves. A piezo buzzer can play a decent role by producing high-pitched tones unpleasant to birds [13]. The buzzer is effective in deterring some birds at a frequency of 1500 Hz compared to other frequencies [11].

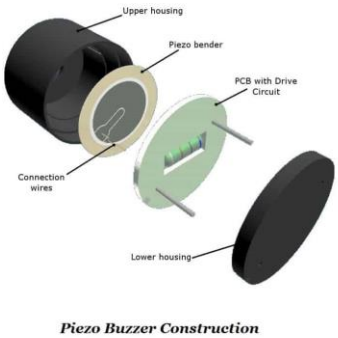


Figure 14: Piezo Buzzer

2.2.8 MP3 Module and Speaker

MP3 module in bird-repellent systems utilizes sound as a deterrent [19]. This module connects to a microcontroller, allowing it to play pre-recorded repellent sounds stored on an SD card. [14]

When the system detects a harmful bird species, the microcontroller selects a sound from a library and plays it through the MP3 speaker. This approach creates an unpredictable auditory environment that confuses birds.

By incorporating machine learning algorithms, the system can analyze audio data in real-time, identifying bird calls and adjusting the repellent sounds accordingly. This adaptability ensures that the bird repellent remains effective over time, providing a responsive and dynamic solution for protecting crops from avian pests [14].

2.2.9 Servo Motor

Servo motors are electromechanical devices that allow precise control of angular or linear position, speed, and acceleration. They operate using a feedback mechanism, allowing them to maintain a specific position based on input commands. This capability makes them ideal for applications requiring accurate movement and control.

Servo motors can effectively simulate human activity, significantly enhancing the scarecrow's ability to deter unwanted birds from crops or other areas [10].

Figure 15: TowerPro SG 90 Micro Servo Motor

2.2.10 Drones

A drone shown in Figure 16 is an unmanned aircraft capable of autonomous flight, operated remotely from the ground using a remote control as the drone's controller [20].

Figure 16: DJI Mini 3 Pro Quadcopter Drone with RC-N1 Remote Controller

Smart drones are an innovative approach to repelling flocks of birds that threaten farmers' crops. The smart drone is equipped with a camera and programmed to detect bird flocks. To deter these flocks, the smart drone is also equipped with a speaker capable of emitting sounds to distress the birds [20]. Additionally, farmers can monitor which rice fields are most frequently visited by bird flocks through a dedicated website [20].

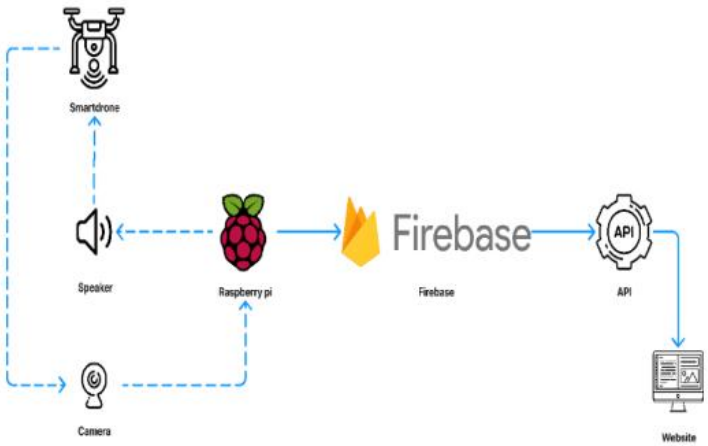


Figure 17: System Design

Hemelrijk's team developed the *RobotFalcon shown below*, a robotic peregrine falcon designed to deter birds at airports [21]. Mimicking a real falcon, it uses small propellers to reach predator-like speeds, scaring birds away. Field tests showed the RobotFalcon outperformed the

conventional drones, keeping birds at bay longer and provoking stronger escape responses without the birds becoming accustomed to it.

Figure 18: RobotFalcon's view as it hunts

2.2.11 Motion Activated Jet Spray

Motion-activated jet sprays are an effective, eco-friendly solution to deter birds. The system uses a motion sensor, typically a Passive Infrared (PIR) sensor, to detect the presence of birds. When movement is detected, the sensor triggers a brief burst of water from a connected jet spray nozzle, startling the bird and prompting it to fly away.

In Figure 19, the jet spray targets the bird with a burst of water, successfully deterring it from the area. This timing is calibrated to provide enough water to disrupt the bird without excessive water use [22].

Figure 19: Jet spray deters the bird with a burst of water

2.2.12 Multi-Actuation Approach

Various bird-deterring methods offer unique advantages, but the most sustainable solutions combine visual, audio, and other techniques. The development of bird-repellent systems seeks to enhance effectiveness by combining the strengths of various deterrent approaches [8]. Figure 20 shows a scarecrow model with sound and motion output.

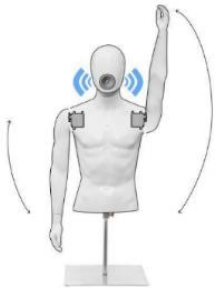


Figure 20: Sound and Motion Output Model of Scarecrow

2.3 Comparative Analysis of Birds Deterrents Actuation Methods

This analysis compares various bird-repellent methods: Table 1—**drones, water jet sprays, laser actuators, ultrasonic deters, and audio deters**—by evaluating their effectiveness, efficiency, operational complexity, scalability, and potential for harm.

Smart drones, while promising as a bird deterrent, face several challenges that impact their effectiveness and feasibility [21]. Its high initial investment and operational costs arise from advanced technology and the need for skilled operators, complicating implementation. Birds may acclimatize with time, reducing the effectiveness of solutions. Furthermore, if laser actuators are employed, there is a risk of harm to birds' eyes if not used carefully, necessitating stringent safety measures to ensure the humane treatment of wildlife [15].

2.3.1 Birds Deterrents Comparative Analysis Table

Actuation Method	Effectiveness	Operational Complexity	Scalability	Humane Impact	Long-Term Response of Birds
Smart Predator Drone	High: Clears fields of birds faster and more effectively than drones.	High: Requires skilled operators with falconry experience.	Moderate: Limited by the need for trained operators.	Mostly humane	There is no acclimatization currently; it mimics a real falcon.
Drones	High: Strong visual and audible disturbance	High: Requires skilled operation and regular maintenance.	Moderate: Limited by cost and operational demands.	Mostly humane	Birds may habituate over time, leading to decreased effectiveness.
Water Jet Spray	High: Birds dislike unexpected sprays and associate them with danger.	Low: Once installed, it operates with minimal maintenance.	High: Scalable over large areas with sprinkler extensions.	Humane: Safe and non-harmful; causes only temporary discomfort.	Birds generally do not habituate significantly to water sprays; they remain uncomfortable.
Laser Deterrents	Moderate: Effective in low-light conditions; mimics threatening visuals.	Low: It can be automated with little maintenance.	Moderate: Effective in smaller zones; additional lasers are required for larger areas.	Humane if low-power: Mimics light flashes without causing physical harm.	Birds may initially react strongly to lasers, but habituation could occur.
Ultrasonic Deterrents	Mixed: Effective for some species; certain birds may adapt or be unaffected.	Low: Simple installation and minimal maintenance.	Moderate to High: Effective in enclosed/semi-enclosed spaces.	Humane: Emits non-harmful sound within bird hearing range.	Varying responses; some species may adapt and become desensitized.
Audio Deterrents	Moderate: Effective short-term; birds may habituate to repeated sounds.	Low: Simple to install; varied sounds improve effectiveness.	High: It covers large areas with amplified sounds; and requires thoughtful deployment.	Generally humane: Predator sounds cause alarm without physical threat.	Birds often habituate to repeated sounds, leading to reduced effectiveness.

Table 1: Birds Repellent Comparative Analysis

2.4 Top 3 Bird Deterrent Options

This ranking evaluates three bird deterrent methods based on effectiveness, cost, humane approach, and scalability. It covers visual and audio deterrents, water jet sprays, and their potential synergy with drones.

2.4.1 Visual Deterrents (Laser and Servo Motors)

i **Effectiveness**: Mimics predator behavior to scare birds. Sudden movements and light elicit escape responses.
ii **Cost**: Moderate initial setup and maintenance expenses.
iii **Humane Approach**: Non-lethal, causing no physical harm.
iv **Scalability**: Can be scaled across large areas.

Summary: Visual deterrents are effective but research needs to be done to determine if lasers affect birds' vision [16].

2.4.2 Bioacoustics (Audio Stimuli)

i **Effectiveness**: Uses distress calls and predator sounds. Habituation can reduce effectiveness.
ii **Cost**: Low-cost installation and maintenance.
iii **Humane Approach**: Non-invasive and humane.
iv **Scalability**: Easy to deploy across large areas.

Summary: Audio deterrents are affordable and humane, effective when tailored for specific bird species.

2.4.3 Water Jet Spray System

i **Effectiveness**: Startles birds with sudden movement and noise. Periodic adjustments are needed.
ii **Cost**: Moderate setup costs, low ongoing expenses.
iii **Humane Approach**: Non-lethal, causing no harm at low pressure.

iv **Scalability**: Simple to install and expand.

Summary: Water jet sprays are practical and cost-effective for diverse agricultural areas.

2.4.4 Synergy with Drones

Integrating a drone system, such as the RobotFalcon, with these methods enhances overall efficacy. The RobotFalcon mimics a natural predator, and when combined with audio and visual deterrents, it promises a comprehensive bird management solution.

2.5 Factors affecting effectiveness

Several factors influence the effectiveness of electronic bird deterrent systems in rice farms:

i Scale of Implementation: Larger rice farms require systems with greater coverage or multiple coordinated units.

ii Bird Species: Different bird species respond differently to deterrent methods. For example, some birds are more sensitive to specific frequencies or more easily frightened by laser patterns.

iii Environmental Conditions: Weather, time of day, and seasonal variations affect system performance. Solar-powered systems may have reduced effectiveness during extended cloudy periods, while camera-based systems may struggle in fog or heavy rain. Laser is inefficient when there's a lot of ambient light.

iv Implementation Quality: Proper installation, calibration, and maintenance significantly impact effectiveness. Optimal positioning of sensors and deterrent mechanisms is crucial.

v Integration with Other Methods: Electronic deterrents often achieve maximum effectiveness when integrated with complementary approaches, such as habitat modification or strategic timing of planting

2.6 Cost-Effectiveness Considerations

When evaluating the cost-effectiveness of electronic bird deterrent systems, several factors must be considered:

i Initial Investment vs. Crop Protection: Higher-cost systems like laser-based deterrents may be justified for large farms where potential crop losses are substantial.

ii Operational Costs: Solar-powered systems have minimal operational costs, while grid-powered systems incur ongoing electricity expenses.

iii Maintenance Requirements: More complex systems typically require more frequent maintenance and technical expertise.

iv Lifespan and Durability: Weather-resistant systems with quality components offer better long-term value despite higher initial costs.

v Coverage Area: Cost per protected acre varies significantly between systems, with laser-based and IoT systems typically offering the most economical large-area coverage

3 DESIGN METHODOLOGY

The bird-repellent system combines visual and auditory deterrents to create an environment that birds find uninviting. The **visual component** includes **lasers** and **servo-controlled movement** to simulate human-like presence and visual disturbance. The **bioacoustics component** uses **distress and predator calls** to trigger an instinctive avoidance response in birds.

The AI-powered bird detection system, including camera integration and deep learning algorithms, is developed separately by Carnegie Mellon University. This project focuses on the actuation subsystem (lasers, servos, audio), which receives detection signals from CMU's module.

3.1 Visual Repellants

Lasers will produce sweeping light patterns across the field to create an uncomfortable bird environment. A Class II **Diode Laser** is a low-power device with wavelengths around 532nm (green) or 650nm (red) typically used, as they are highly effective in deterring birds [16].

The FDA recognizes four major hazard classes (I to IV) of lasers. The higher the class, the more powerful the laser is and the greater the potential to pose serious injury if used improperly.

The proposed laser-based system will operate as follows:

1. The camera continuously monitors the protected area
2. The deep learning network processes images to identify birds in real-time
3. When birds are detected, the system calculates their position
4. The laser targeting mechanism directs the beam toward the detected birds
5. Birds perceive the approaching laser as a threat and flee
6. The system continues monitoring and targeting as needed

The precision targeting of the system prevents birds from becoming habituated to the laser, maintaining effectiveness over extended periods. Figure 21 shows a laser-based deterrent system.

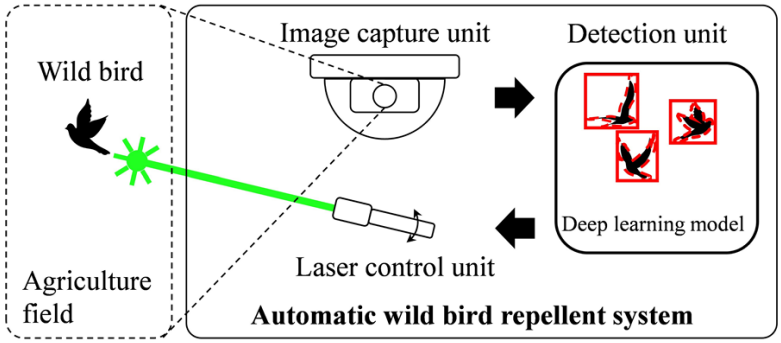


Figure 21: A laser repellent system

Laser- based systems show highest compatibility with real-time bird tracking, enabling precision deterrence.

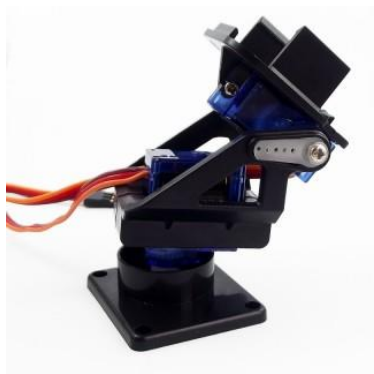


Figure 22:Pan and Tilt with Micro Servo

3.2 Bioacoustics Repellants

Pre-recorded bird distress calls, predator calls (hawk, falcon, owl), and loud noises will play at random intervals to deter birds by triggering their survival instincts. This will be achieved by an MP3 Player Module, like the DFPlayer Mini, and an SD CARD, as shown in Figure 23: for storing and playing audio files with minimal power consumption.

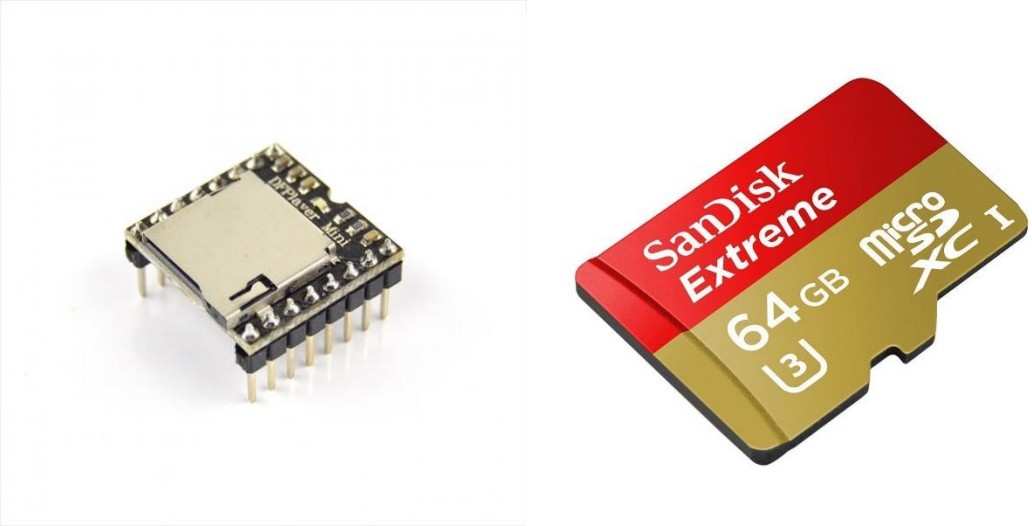


Figure 23: The DFPlayer Mini MP3 Player and a SanDisk 64 GB SD Card

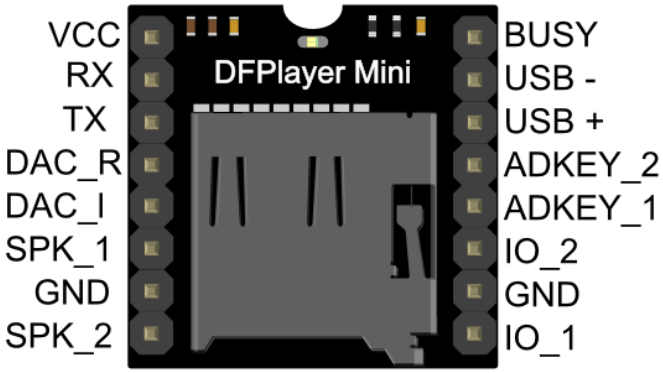


Figure 24:DFPlayer Mini Pin Map

Additionally, durable MP3 speakers, such as the one in Figure 25, are capable of considerable audio output to cover large fields. Waterproofing is important if installed outdoors.

Figure 25: A 2 Watts 8 Ohm MP3 Speaker

3.3 The Controller

The microcontroller will be the "brain" for synchronizing lasers, motor movements, and sound playback.

3.3.1 Microcontroller Requirements for Bird Repellent System

i **Processing Power**: Handles sensors, actuators, and timing.

ii **I/O and Peripherals**: Adequate GPIO pins and SPI/I2C/UART interfaces.

iii **Connectivity**: LoRa, ZigBee, or Bluetooth for remote monitoring/control.

iv **Power Efficiency**: Supports low-power modes for solar-powered systems.

v **Memory**: Sufficient for code storage and algorithms.

vi **Ease of Use**: Simple programming, strong community support, and libraries.

vii **Cost-Effectiveness**: Balances features and affordability for scalability.

Figure 26: ESP-WROOM-32 Dual Core CPU with Low Power Consumption

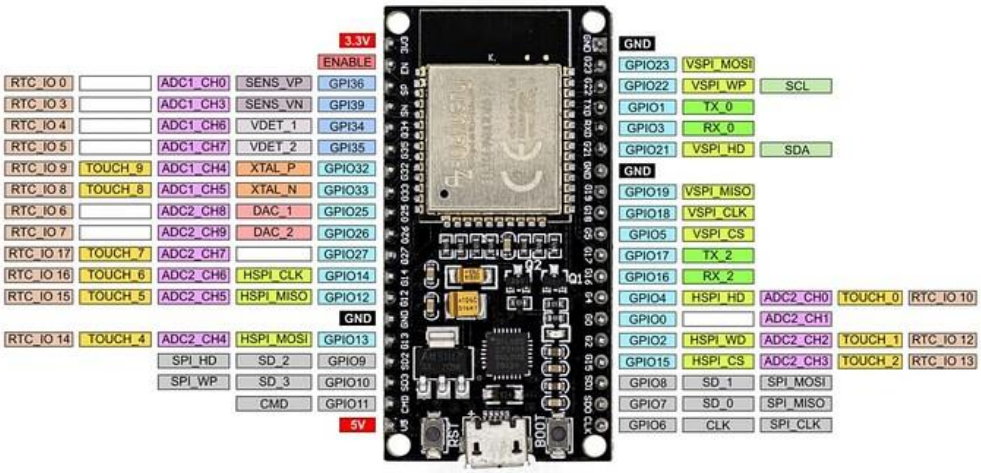


Figure 27: ESP-WROOM-32 pinout

3.4 Bird Repellent System Block Diagram

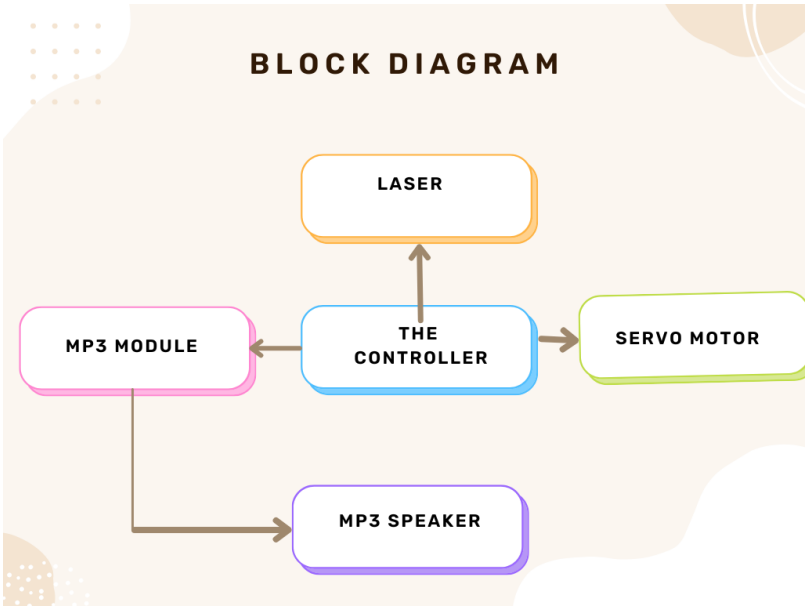


Figure 28: Block Diagram Representation of the System

The actuation subsystem is designed to interface seamlessly with the AI-powered bird detection module developed by Carnegie Mellon University. The detection module performs real-time image analysis using deep learning and outputs a digital signal upon detecting a bird within the protected field of view. The actuation subsystem is expandable to receive coordinate data via UART or I²C, allowing precise control of the laser's targeting mechanism through servo-driven pan and tilt systems. This modular approach enables flexibility in future upgrades—such as using LoRa or Zigbee for remote communication—and allows separation between the sensing and actuation units to cover larger fields.

3.5 Audio Stimuli

Predator and distress calls, including hawk and falcon sounds, were sourced from **birdweather.com,** a living library of bird vocalization. Link: https://app.birdweather.com/login

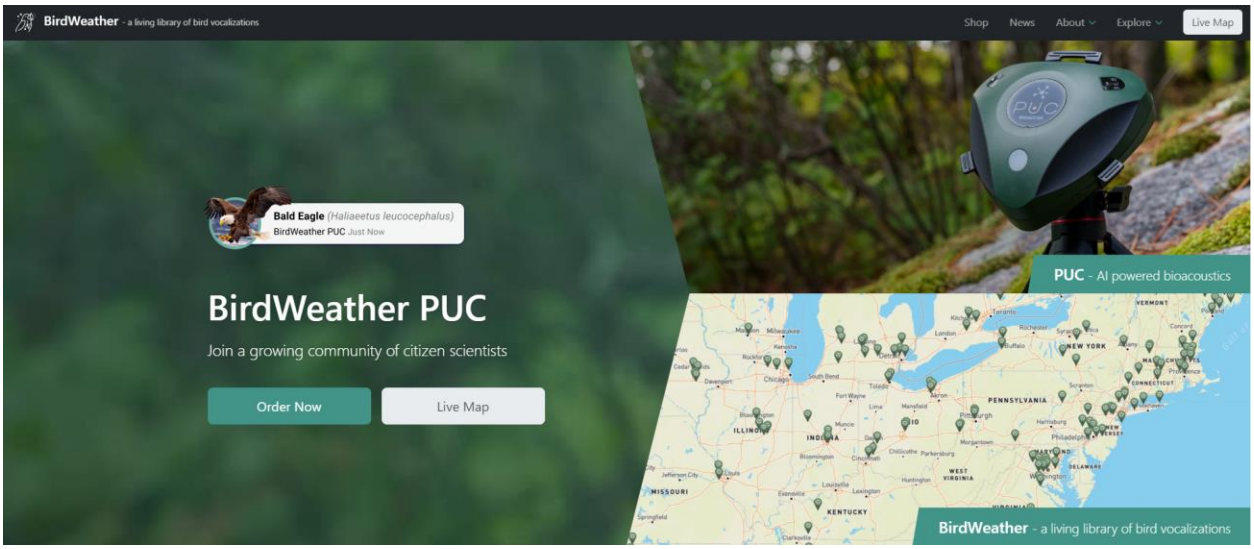


Figure 29: BirdWeather - a living library of bird vocalization

BirdWeather is a pioneering visualization platform that harnesses the **BirdNET** artificial neural network to monitor bird vocalizations globally through over 2000 active audio stations. The platform is a comprehensive, evolving digital repository of bird sounds. BirdWeather facilitates easy recording and automatic cloud processing of bird sounds, aiding conservation efforts and ecological studies.

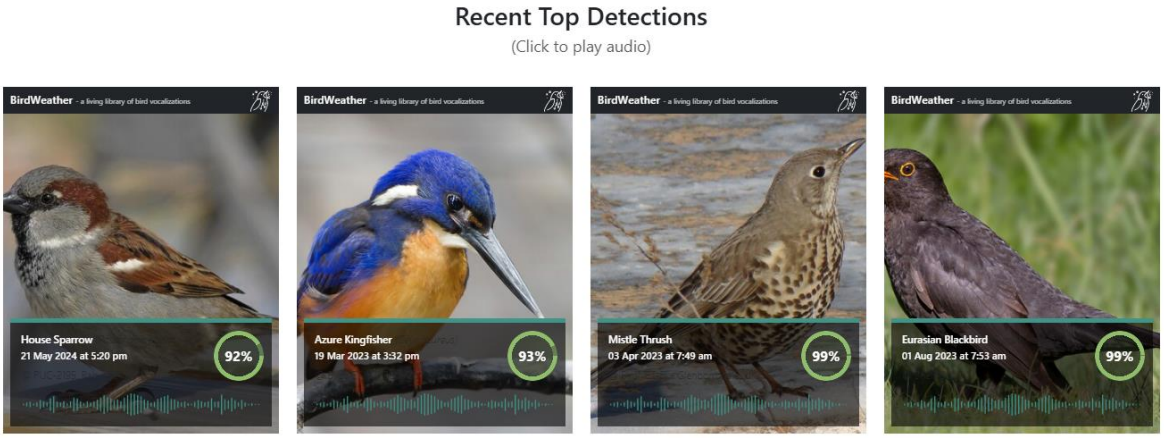


Figure 30: Some Natural Bird' Sounds in BirdWeather

By analyzing bird population behaviors and changes, BirdWeather provides valuable insights into environmental shifts, engaging the public in global citizen science initiatives through its web dashboard and mobile app [23].

The selected sounds were tested on avian species around the home environment, including crows, robins, pigeons, and sparrows. Most birds' auditory ranges overlap with humans (1 kHz to 4 kHz).

While predator calls triggered a moderate avoidance response, their effectiveness was limited as birds frequently searched for accompanying visual cues, diminishing the overall deterrence.

3.6 Visual Stimuli

A **green diode laser** mounted on a servomechanism will produce randomized sweeping light patterns to create visual stimuli. The system will integrate synchronized audio and visual stimuli to enhance bird deterrence. The laser-based system is inspired by methodologies from the scientific report **"Automatic wild bird repellent system that is based on deep-learning-based wild bird detection and integrated with a laser rotation mechanism."** [16]

Figure 31: Laser Bird Control Device with Laser Light to Repel Bird

3.7 System Flowchart

The microcontroller will implement dynamic servo movements and randomization of sound playback. The timing and synchronization will avoid predictable patterns that birds might adapt to.

The algorithm will synchronize laser, motor-driven actions, and audio stimuli to create a dynamic and unpredictable environment that effectively deters birds.

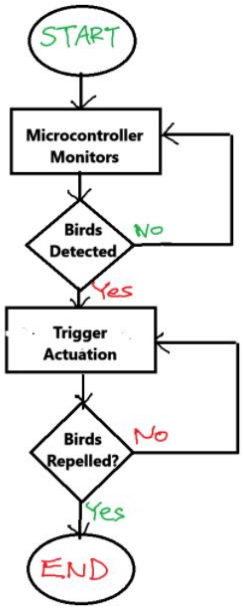


Figure 32: General flowchart of the bird deterrent algorithm

3.8 Simulation on Wokwi

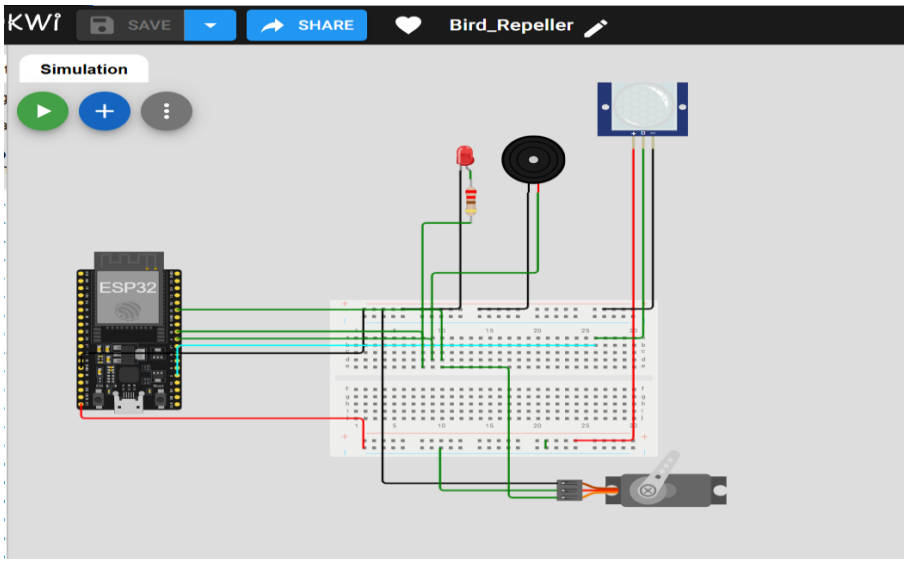


Figure 33: Simulation on Wokwi (Web-Based Simulator)

The following components were simulated using **Wokwi (ESP32-based simulation)**:

i **PIR Sensor (Motion Detection)** – Detects motion events, simulating bird presence.

ii **Servo Motor (Motion driven Deterrent)** – Sweeps back and forth in response to motion.

iii **Buzzer (Audio Deterrent and sound playback)** – Generates audible frequency (1KHz-4KHz)

iv **LED Indicator** – Turns ON when motion is detected, OFF otherwise. (LASER's replacement in simulation)

3.9 Prototype

3.9.1 Breadboard

Initial testing of core components (MP3 module, servos, laser). The DFPlayer Mini module communicates with the microcontroller via UART, with its RX pin connected to the TX pin of the ESP32 and its TX pin connected to the RX pin of the microcontroller. Two servo motors are controlled using PWM signals from GPIO pins 26 and 27. Additionally, four slide switches are connected to GPIO pins 2, 5, 18, and 19, enabling manual control and mode selection. These connections form the core of the prototype's wiring, facilitating coordinated actuation of the deterrent system.

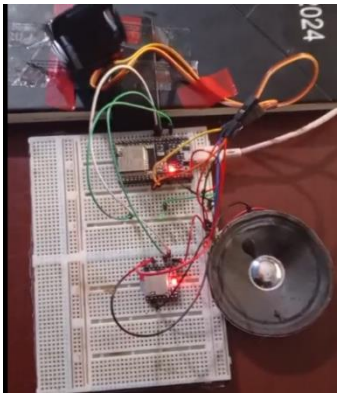


Figure 34: Breadboard version

3.9.2 Electrical Hardware Design

The prototype PCB was designed using **EasyEDA**, a robust electronic design automation tool. Figure 35 shows the complete Actuation Board Schematic, which integrates several key components essential for the bird repellent system.

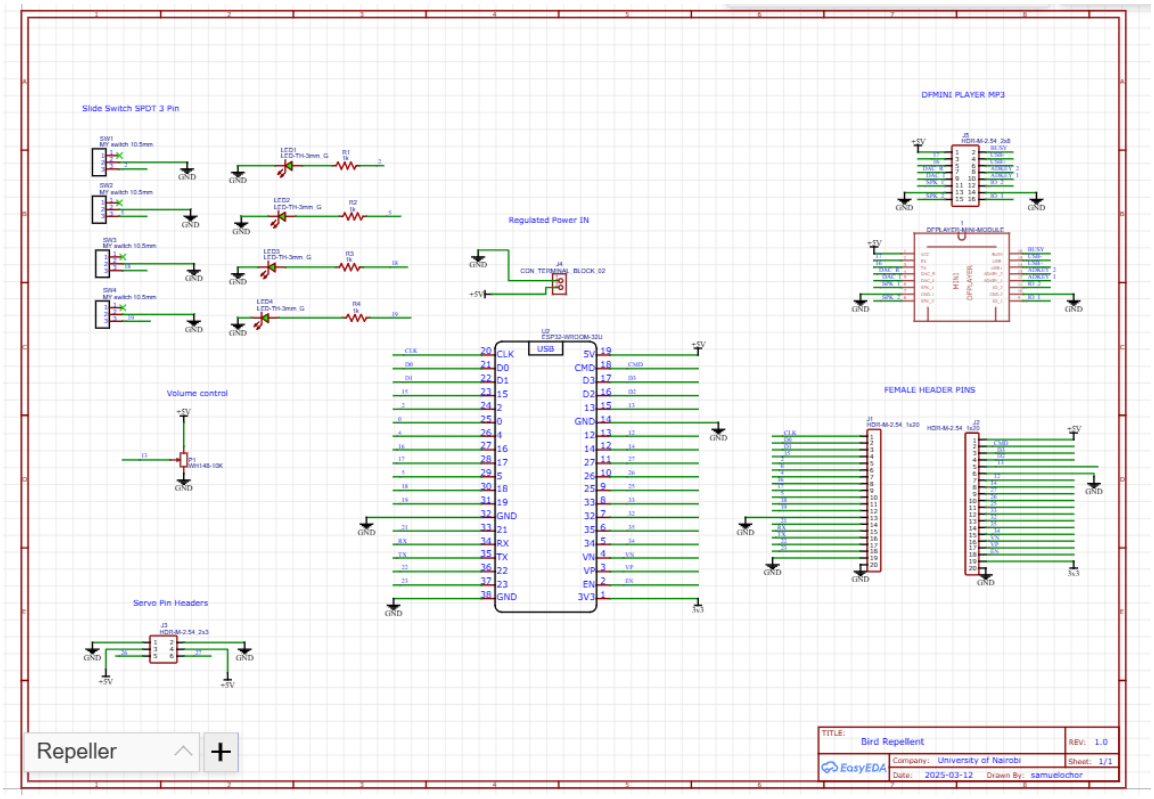


Figure 35: Actuation Board Schematic

3.9.3 Key components

i **Slide Switch SPDT**: For manual switching between different states (Test mode, Idle mode and Active mode)

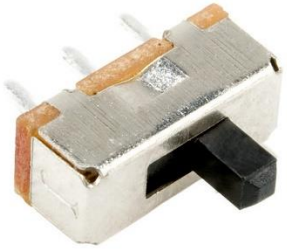


Figure 36: Slide Switch SPDT

ii **LED indicators**: To show system status and activation states.

iii **Volume control**: The potentiometer in this setup serves as a manual control for adjusting the volume of the sound-based deterrent. It operates as a voltage divider, where the voltage across the resistive track is divided based on the position of the wiper. As the wiper moves, it outputs a voltage that is proportional to its position along the track—ranging from 0V to 5V. This analog voltage is then read by the ESP32 microcontroller through an analog input pin. Within the microcontroller, this input is mapped to a desired output volume range—from 0 to 30—using a mapping function. This mapped value is used to adjust the volume output of the DFPlayer Mini module, which controls the loudness of the audio deterrent. This approach allows the user to vary the intensity of the audio repellent based on the surrounding environment or bird response, adding flexibility and real-time control to the system.

iv **Servo motor pin headers**: To connect the motion-based repellent system.

v **MP3 player module (DFPlayer Mini)**: To play predator or distress sounds.

vi **ESP32 microcontroller**: For processing input data and controlling outputs.

vii **Power Input**: Stable 5V supply for all modules.

viii **Connectivity headers**: For expansion and integration of additional modules. The headers will allow integration with CMU's detection module. Expected input: digital triggers (e.g., 'bird detected' signal) and coordinates (for laser targeting).

4 IMPLEMENTATION

4.1 Electrical Hardware Implementation

The PCB was fabricated using JLCPCB services and populated by hand-soldering the components. The fabricated board is shown in Figure 39.

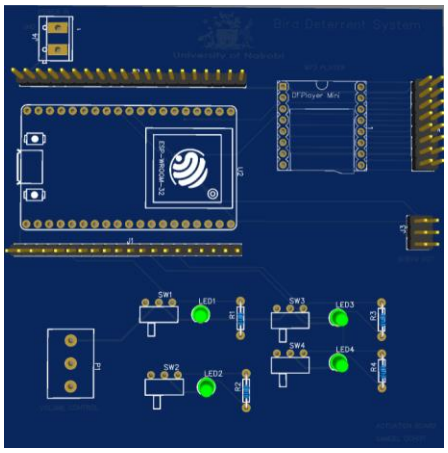


Figure 37: The Actuation PCB Top View

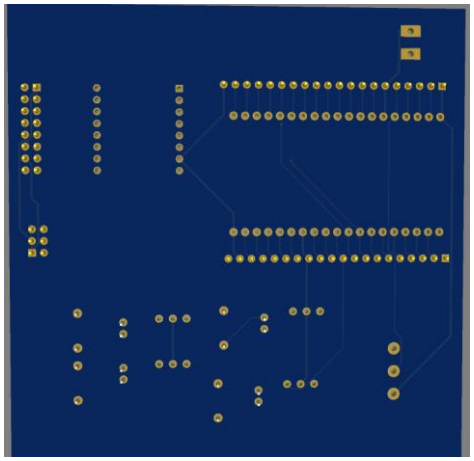


Figure 38: The Actuation PCB Bottom View

The Prototype integrates key features for the automated bird repellent system, combining audio playback, physical actuation, visual deterrence, and manual control capabilities.

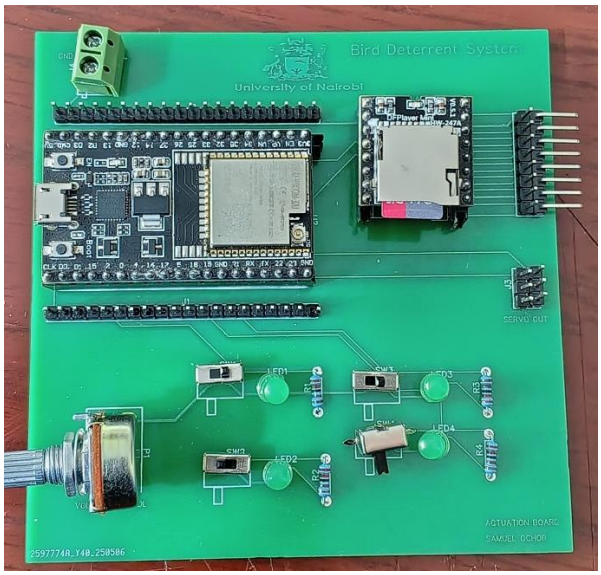


Figure 39: The Actuation PCB

4.1.1 Core Features

i ESP32-WROOM-32 as the main controller, with accessible (broken out) GPIO header (J1) for future expansion.

ii DFPlayer Mini MP3 module socket for acoustic deterrent, paired with a volume control potentiometer. MP3 file “020” activates only when the system is in “Test Mode”.

Figure 40: 20 MP3 sound files stored on the SD card for playback when birds are detected.

iii Servo output header (“SERVO OUT”).

iv Slide switches (SW1–SW4) for manual mode selection.

v LEDs with current-limiting resistors for visual feedback on system status.

vi Power input terminal (2-pin) to supply the board and peripherals.

vii Clear silkscreen labels for all modules (e.g., “VOLUME CONTROL”, “MP3 PLAYER”).

viii Modular layout for easy debugging and assembly.

ix Expansion-friendly via extra pin headers.

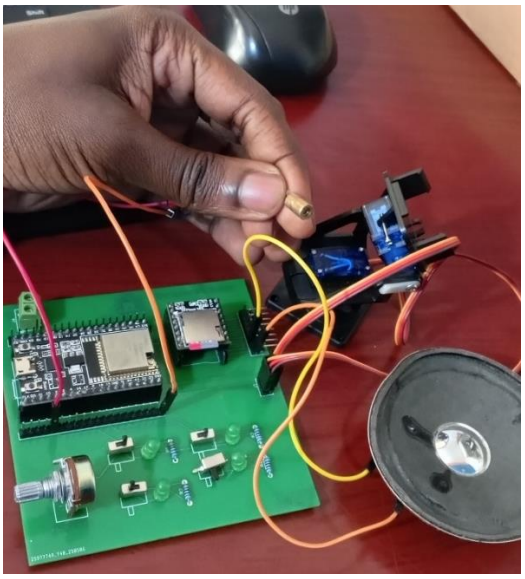


Figure 41: Prototype

4.2 Software Implementation

Firmware was developed in the Arduino IDE environment, utilizing libraries for servo control, MP3 playback. The detection logic triggers the activation of laser and sound modules, introducing random variations in the actuation patterns to prevent birds from adapting.

4.2.1 Preliminaries

The following software was installed before the start of the implementation:

i Arduino IDE

ii CP210x USB to UART Bridge Virtual COM Port (VCP) drivers

4.2.2 Installing CP210x USB to UART Bridge VCP Drivers (Windows)

The CP210x USB to UART Bridge Virtual COM Port (VCP) drivers were essential for establishing communication between the computer and the ESP board, which uses the CP210x USB-to-serial bridge chip for data transfer. These drivers allowed the operating system to recognize and interface with the ESP board as a standard serial port, facilitating seamless uploading of firmware and serial monitoring during development.

The drivers were downloaded from the official Silicon Labs website and installed according to the operating system's instructions. Once installed, the ESP board appeared as a COM port in the Device Manager (Windows). The assigned COM port number was then used in the Arduino IDE to upload code and interact with the ESP board.

4.2.3 Arduino IDE

The Arduino IDE was installed to provide a user-friendly environment for writing, compiling, and uploading code to the ESP board. The IDE was downloaded from the official Arduino website and installed following the platform-specific instructions.

Within the Arduino IDE, the **Board Preferences** were configured to include support for the ESP32 boards. This was achieved by adding the ESP32 Board URL to the **Additional Boards Manager URLs** in the **Preferences** menu. The ESP32 Board definitions were then installed using the **Boards Manager**, enabling the IDE to recognize and compile code for the ESP32 board.

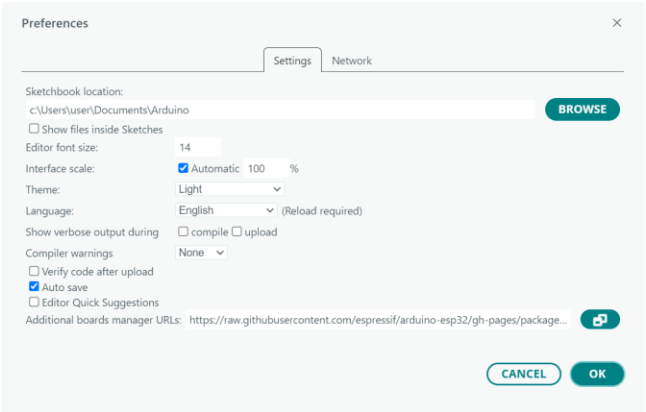
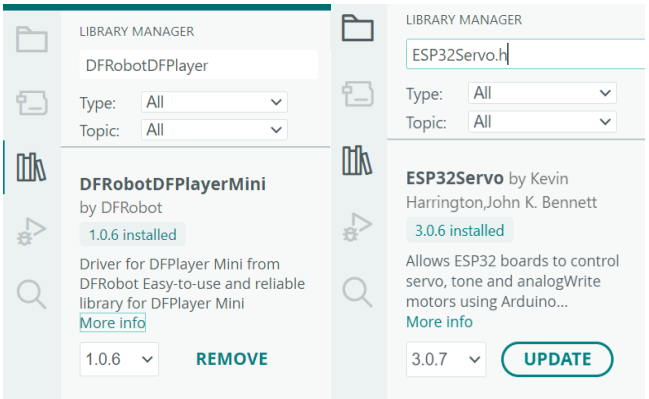


Figure 42: Additional Boards Manager

Additionally, the necessary libraries were installed to support the project's functionality. The **ESP32Servo** library was included to control the servo motor, and the **DFPlayerControl** library was added to manage audio playback from the MP3 module. These libraries were installed through the **Library Manager** in the Arduino IDE and were included in the sketch.



For readability and ease of management, the software was split into .h header files and .cpp source files. This modular approach allowed for better organization, improved debugging, and simplified updates to individual parts of the system Figure 43.

4.3 Program Implementation

The program was implemented in the Arduino IDE to control the bird-repellent system using the ESP32 microcontroller. The code was modularly organized by splitting it into multiple .h header files and .cpp source files to ensure easy management, readability, and debugging.

The main program included essential libraries and custom header files. When **system Status** was 1, the deterrent system was active, engaging the servo-based visual deterrent and the audio module for random playback of distress or predator sounds. When **system Status** was 0, the system entered **Test Mode** if **test Status** was 1, triggering deterrent functions for testing and verification.

The code also provided real-time serial control of deterrent mechanisms and allowed easy adjustments and testing of the system's functionality.

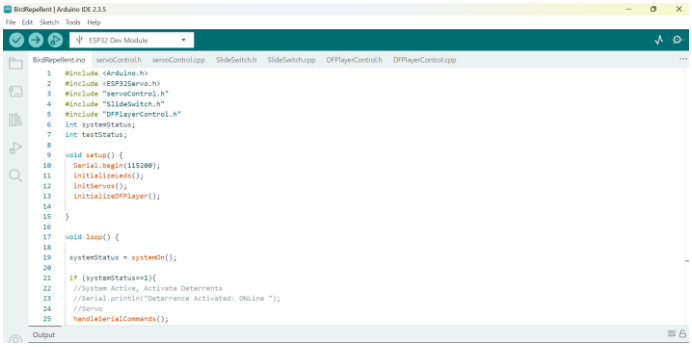


Figure 43:Arduino Code Split into .h and .cpp Files for Modular Design

BirdRepellent	5/29/2025 2:25 PM	INO File	1 KB
DFPlayerControl	5/29/2025 2:12 PM	C++ Source File	4 KB
DFPlayerControl	5/29/2025 2:13 PM	C Header Source File	1 KB
servoControl	5/29/2025 2:28 PM	C++ Source File	4 KB
servoControl	5/29/2025 2:25 PM	C Header Source File	1 KB
SlideSwitch	5/29/2025 9:46 AM	C++ Source File	3 KB
SlideSwitch	5/29/2025 9:38 AM	C Header Source File	1 KB

Figure 44: Bird Repellent Project Files

5 RESULTS

The prototype system integrated key deterrent components—laser activation on trigger, dynamic servo movement, and randomized audio playback—into a multi-modal bird-repellent system. Bench tests confirmed the system's responsiveness and coordination between audio-visual deterrents.

These preliminary tests demonstrated the feasibility of the system's core functions.

```
 Enter angles as: pitch,yaw
------------------------------
Attaching servoPitch to pin 26
Initializing DFPlayer Mini...
DFPlayer Mini ready.
Volume adjusted via potentiometer: 30
Playing track: 2
Playing track: 14
Moved to: Pitch=90, Yaw=90
Playing track: 9
Playing track: 15
Moved to: Pitch=0, Yaw=0
Playing track: 2
Volume set via serial: 30
Playing track: 10
Playing track: 7
Playing track 13
Playing track: 15
Playing track: 18
Playing track: 15
Moved to: Pitch=180, Yaw=0
Playing track: 16
    The servo is in test mode
```

Figure 45: Arduino IDE's Serial Monitor

While the bench-level tests validated core functionalities, a **quantitative evaluation of real-world deterrence effectiveness** is essential for deployment in rice field environments. The following framework is inspired by methodologies from the scientific report **"Automatic wild bird repellent system that is based on deep-learning-based wild bird detection and integrated with a laser rotation mechanism."** [16]

To objectively quantify effectiveness, two key indicators would be needed. The **first indicator** that would be used to quantify the deterrent system's effectiveness is the **daily bird repulsion**

rate (BRR_d), defined in Eq. (3). In this approach, N_{on} and N_{off} would represent the number of wild birds detected in each image captured during one-hour intervals when the system is **active** (on) and **inactive** (off), respectively. The subscript (i) denotes the hour of observation [16].

The summed number of wild birds observed throughout the day would estimate the scenario of bird presence in the field. Comparing the number of birds during active and inactive periods would then allow for the quantification of the system's repulsive effect. To ensure fair comparisons, the same counting and calculation method would be consistently applied across all time intervals. Typically, data would be captured from **06:00 to 18:00** in a single day, with the system operating **6 hours on** and **6 hours off** [16].

In addition to daily deterrence, the **hourly bird repulsion rate (BRR_h)**, defined in Eq. (4), would also be used. This metric would compare the number of birds in each active hour ($N_{on,i}$)with the average number of birds in the **adjacent inactive hours** ($N_{off,i-1}$)and ($N_{off,i+1}$). BRR_h would therefore measure the hourly repulsion effectiveness of the system, taking natural variations in bird activity into account [16].

$$BRR_d = \frac{\sum_{In\ one\ day} N_{off,i+1} - \sum_{In\ one\ day} N_{on,i}}{\sum_{In\ one\ day} N_{off,i+1}} \times 100\% \quad (3)$$

$$BRR_h = \frac{\frac{(N_{off,i-1}+N_{off,i+1})}{2} - N_{on,i}}{\frac{(N_{off,i-1}+N_{off,i+1})}{2}} \times 100\% \quad (4)$$

6 RECOMMENDATIONS

The following recommendations are proposed to enhance system effectiveness, scalability, and future development:

1. Develop a coordinate transformation framework to support non-co-located setups of the detection and deterrence subsystems. Collaboration with geospatial experts is advised.
2. Real-world testing to improve actuator precision and test overall deterrence effectiveness.
3. It is highly recommended to engage local farmers through training sessions on operating, maintaining, and troubleshooting the system. Their feedback should also be incorporated for practical improvements and optimization of the system.
4. A thorough study of the system's ecological footprint is advised before large-scale deployment. Particular attention should be given to ensuring both target and non-target species are not adversely affected.
5. Integrate the actuation, power, communication, and potentially sensing subsystems into a single compact unit to enhance ease of deployment, reduce costs, minimize physical footprint, and improve overall system reliability.
6. Conduct periodic laser safety assessments and consider compliance with IEC laser standards for future scaled deployment.

7 CONCLUSION

This project addressed the design and implementation of an actuation system for a bird repellent device, targeting effective deterrence of birds in a rice farm environment. By thoroughly reviewing existing deterrent mechanisms, it established a solid foundation for the hardware design and control algorithms that enable real-time activation of the actuators based on sensor feedback from the bird detection subsystem.

Beyond the immediate technical scope, this work contributes to the broader Sustainable Development Goals (SDGs), particularly **SDG 2 (Zero Hunger)** and **SDG 12 (Responsible Consumption and Production)**. By mitigating crop losses due to bird interference, the work promotes food security and agricultural productivity in a sustainable manner. Overall, this project not only contributes to the field of smart farming but also highlights the importance of engineering solutions in addressing pressing global challenges.

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9 APPENDICES

9.1 APPENDIX 1: Bill of Quantities

Component	SKU	Quantity	Cost (KES)	**Total (KES)**
Micro Servo 9g TS90 Degree Digital Servo Motor	PEMSM43	2	300.0	600
Micro Servo bracket Pan/Tilt Kit	PESBP77	1	200.0	200
Magnetic speaker 0.5W 8R (18MM)	PEYHD32	1	100.0	100
Mini MP3 Player Module TF Card	PEMTF44	2	300.0	600
SanDisk Ultra 32GB Class 10 Micro SD card	PECBN49	1	1300.0	1300
Breadboard 830 Point Solderless MB102	PEPCB58	1	20.0	20
65pcs Jump Wire Cables Male to Male	PEDEC28	1	170.0	170
650nm 5mW Adjustable Laser Dot Module	PERYU44	1	50.0	50
Buzzer Piezoelectric Active	PEMAB77	1	20.0	20
ESP32S DevKIT WIFI + BLE Module (30Pin)	PEWBA82	1	1400.0	1400
Slide Switch		4	20	80
Header Pins		5	30	150
Diffused 3mm Green Led + Red Led	PELED81G	4	5.0	20
Total				**4890.00**

9.2 Appendix 2: BirdRepellent.ino

```
//Modularized software design

//Main file BirdRepellent.ino

#include <Arduino.h>
#include <ESP32Servo.h>
#include "servoControl.h"
#include "SlideSwitch.h"
#include "DFPlayerControl.h"
int systemStatus;
int testStatus;

void setup() {
  Serial.begin(115200);
  initializeLeds();
  initServos();
  initializeDFPlayer();

}

void loop() {

 systemStatus = systemOn();

 if (systemStatus==1){
  //System Active, Activate Deterrents
  //Serial.println("Deterrence Activated: ONLine ");
  //Servo
  handleSerialCommands();

  //DF Player Control
  //processSerialCommands();
  checkPotVolume();
  handleRandomPlayback();
```

```
 }else if (systemStatus==0 ){
  testStatus = testMode();
  if (testStatus==1){
    //Test Deterrents in Test Mode
    //Serial.println("System in Test Mode");
    test_system();

  }else{
    //Serial.println("System OFFLine ");}}}
```

9.3 Appendix 3: servoControl.h

```
#pragma once

void initServos();
void handleSerialCommands();
void updateServos();
void moveServos();
void test_system();
```

9.4 Appendix 4: servoControl.cpp

```
#include "servoControl.h"
#include <ESP32Servo.h>
#include <DFPlayerControl.h>

extern int pitchDir = 2;
extern int yawDir = 4;

Servo servoPitch;
Servo servoYaw;

extern int pitchPos = 90;
extern int yawPos = 90;

void initServos() {
  Serial.println("------------------------------");
  Serial.println("    The servo sketch has started    ");
```

```
  Serial.println(" Enter angles as: pitch,yaw ");
  Serial.println("------------------------------");

  if (!servoPitch.attach(26)) Serial.println("Failed to attach servoPitch to pin 26");
  if (!servoYaw.attach(27))   Serial.println("Failed to attach servoYaw to pin 27");

  servoPitch.write(pitchPos);
  servoYaw.write(yawPos);
}


void handleSerialCommands() {
  if (Serial.available()) {
    String input = Serial.readStringUntil('\n');
    input.trim();

    int commaIndex = input.indexOf(',');

    if (commaIndex > 0) {
      int pitch = input.substring(0, commaIndex).toInt();
      int yaw   = input.substring(commaIndex + 1).toInt();

      pitchPos = constrain(pitch, 0, 180);
      yawPos   = constrain(yaw, 0, 180);

      updateServos();

      Serial.print("Moved to: Pitch=");
      Serial.print(pitchPos);
      Serial.print(", Yaw=");
      Serial.println(yawPos);
    }
  //Handle Sound Playback Serial Commands
    else if (input == "play") {
      playRandomSound();
    } else if (input == "help") {
```

```
      printHelpMenu();
    } else if (input.startsWith("track ")) {
      int track = input.substring(6).toInt();
      if (track > 0 && track < numberSounds) {
        myDFPlayer.play(track);
        Serial.print("Playing track ");
        Serial.println(track);
      } else {
        Serial.println("Invalid track number.");
      }
    } else if (input.startsWith("vol ")) {
      int vol = input.substring(4).toInt();
      vol = constrain(vol, 0, 30);
      myDFPlayer.volume(vol);
      Serial.print("Volume set via serial: ");
      Serial.println(vol);
      lastVolume = vol;
      serialVolumeSet = true;
    } else if (input == "pot vol") {
      serialVolumeSet = false;
      Serial.println("Potentiometer volume control re-enabled.");
    } else {
      Serial.println("Unknown command. Try: play, track <n>, vol <n>, pot vol");
      Serial.println("Invalid input format. Use: pitch,yaw");
    }
  }
}

void updateServos() {
  servoPitch.write(pitchPos);
  servoYaw.write(yawPos);
}

void moveServos() {
  Serial.println("    The servo is in test mode   ");
  pitchPos += pitchDir;
```

```
  // rollPos += rollDir;
  yawPos += yawDir;

  if (pitchPos < 0 || pitchPos > 180) {
    pitchDir = -pitchDir;
    pitchPos = constrain(pitchPos, 0, 180);
  }

  if (yawPos < 0 || yawPos > 180) {
    yawDir = -yawDir;
    yawPos = constrain(yawPos, 0, 180);
  }
  servoPitch.write(pitchPos);
  // servoRoll.write(rollPos);
  servoYaw.write(yawPos);
}

// Timers
unsigned long lastServoMove = 0;
unsigned long lastSoundPlay = 0;
unsigned long lastLaserToggle = 0;
bool laserState = false;

void test_system() {
  unsigned long currentMillis = millis();

  // Move servos every 20ms (smoother motion)
  if (currentMillis - lastServoMove >= 20) {
    moveServos();
    lastServoMove = currentMillis;
  }

  // Play random sound every 5-15 seconds
  if (currentMillis - lastSoundPlay >= random(5000, 15000)) {
    int randomInt = random(1, 4);
    if(randomInt==1){
```

```
    playRandomSound();
   }else{
    myDFPlayer.play(20);
   }
   lastSoundPlay = currentMillis;
  }
}
```

9.5 Appendix 5: slideSwitch.h

```
#ifndef SLIDESWITCH_H
#define SLIDESWITCH_H

// Define the pins connected to the switches and LEDs
const int NUM_SWITCHES = 4;
extern const int swPins[NUM_SWITCHES];
extern const int ledPins[NUM_SWITCHES];

// Function declarations
void initializeLeds();
void updateLeds();
int systemOn();
int testMode();

#endif
```

9.6 Appendix 6: slideSwitch.cpp

```
#include "Arduino.h"
#include "SlideSwitch.h"

// Define the pins connected to the switches and LEDs
const int swPins[NUM_SWITCHES] = {2, 5, 18, 19};   // Pins for switches
const int ledPins[NUM_SWITCHES] = {2, 5, 18, 19};  // Pins for LEDs
int switchState;

void initializeLeds() {
  for (int i = 0; i < NUM_SWITCHES; i++) {
```

```
    pinMode(swPins[i], INPUT_PULLUP);
    pinMode(ledPins[i], OUTPUT);
    digitalWrite(ledPins[i], LOW); // Ensure LEDs are off initially
  }
}

void updateLeds() {
  for (int i = 0; i < NUM_SWITCHES; i++) {
    switchState = digitalRead(swPins[i]);

    if (switchState == LOW) {
      digitalWrite(ledPins[i], HIGH);  // Turn on the LED
      // Serial.print("Switch ");
      // Serial.print(i + 1);
      // Serial.println(" is pressed - LED ON");
    } else {
      digitalWrite(ledPins[i], LOW);   // Turn off the LED
      // Serial.print("Switch ");
      // Serial.print(i + 1);
      // Serial.println(" is open - LED OFF");
    }
  }
}
int systemOn() {

  switchState = digitalRead(swPins[0]);
  digitalWrite(ledPins[0], HIGH);

    if (switchState == LOW) {
      digitalWrite(ledPins[1], LOW);  // Turn on the LED
      //Serial.println("System OFFLine ");
      // Serial.print(i + 1);
      // Serial.println(" is pressed - LED ON");
      return 0;
    } else {
      digitalWrite(ledPins[1], HIGH);   // Turn off the LED
```

```
    //Serial.println("Deterrence Activated: ONLine ");
    return 1;
    // Serial.print(i + 1);
    // Serial.println(" is open - LED OFF");
  }
}
int testMode() {

 switchState = digitalRead(swPins[2]);
 digitalWrite(ledPins[2], HIGH);

  if (switchState == LOW) {
    digitalWrite(ledPins[3], LOW);  // Turn on the LED
    //Serial.println("Test Mode OFF ");
    // Serial.print(i + 1);
    // Serial.println(" is pressed - LED ON");
    return 0;
  } else {
    digitalWrite(ledPins[3], HIGH);   // Turn off the LED
    //Serial.println("System in Test Mode");
    return 1;
    // Serial.print(i + 1);
    // Serial.println(" is open - LED OFF");
  }
}
```

9.7 Appendix 7: DFPlayerControl.h

```
#ifndef DFPLAYERCONTROL_H
#define DFPLAYERCONTROL_H

#include <Arduino.h>
#include <DFRobotDFPlayerMini.h>

// HardwareSerial for DFPlayer
extern HardwareSerial mySerial;
```

```
extern DFRobotDFPlayerMini myDFPlayer;

// Pins and configuration
extern const int potPin;
extern int lastVolume;
extern int numberSounds;
extern bool serialVolumeSet;

// Random playback
extern unsigned long lastSoundTime;
extern unsigned long soundInterval;

// Function declarations
void initializeDFPlayer();
void playRandomSound();
void printHelpMenu();
void checkPotVolume();
//void processSerialCommands();
void handleRandomPlayback();

#endif
```

9.8 Appendix 8: DFPlayerControl.cpp

```
#include "DFPlayerControl.h"
#include "servoControl.h"

// Initialize shared variables
HardwareSerial mySerial(2); // UART2: RX=GPIO16, TX=GPIO17
DFRobotDFPlayerMini myDFPlayer;

const int potPin = 13;
int lastVolume = -1;
```

```
int numberSounds = 19;
bool serialVolumeSet = false;

unsigned long lastSoundTime = 0;
unsigned long soundInterval = 0;

void initializeDFPlayer() {
  //Serial.begin(115200);
  mySerial.begin(9600, SERIAL_8N1, 16, 17);

  Serial.println("Initializing DFPlayer Mini...");
  if (!myDFPlayer.begin(mySerial)) {
    Serial.println("DFPlayer Mini not detected!");
    while (true);
  }

  Serial.println("DFPlayer Mini ready.");
  myDFPlayer.volume(30); // Default startup volume Max
}

void playRandomSound() {
  int randomTrack = random(1, numberSounds);
  Serial.print("Playing track: ");
  Serial.println(randomTrack);
  myDFPlayer.play(randomTrack);
}

void printHelpMenu() {
  Serial.println("\n=== DFPlayer Command Menu ===");
  Serial.println("help        - Show this help menu");
  Serial.println("play        - Play a random sound");
  Serial.println("track <n>    - Play specific track number <n>");
  Serial.println("vol <n>      - Set volume (0 - 30)");
  Serial.println("pot vol      - Re-enable potentiometer volume control");
  Serial.println("==============================\n");
}
```

```
void checkPotVolume() {
  if (serialVolumeSet) return;

  int potValue = analogRead(potPin);
  int volume = map(potValue, 0, 4095, 0, 30);

  if (abs(volume - lastVolume) >= 2) {
    myDFPlayer.volume(volume);
    Serial.print("Volume adjusted via potentiometer: ");
    Serial.println(volume);
    lastVolume = volume;
  }
}

void handleRandomPlayback() {
  if (millis() - lastSoundTime > soundInterval) {
    playRandomSound();
    lastSoundTime = millis();    soundInterval = random(5000, 7000); // Next sound in 5–7 seconds
```